

Partitioned Information-Budgeted Prompt Compression: A Generic Optimization Model for Selecting Compatible Compressors under Context Constraints

A proposal grounded in task-oriented informational densification

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Abstract

Long prompts are increasingly used to carry instructions, demonstrations, retrieved documents, dialogue history, and task-specific evidence into large language models (LLMs). This creates computational, financial, and epistemic pressure: longer inputs increase latency and cost, occupy scarce context-window capacity, dilute key information, and may intensify position bias or hallucination. Building on recent work on hard prompt compression, soft prompt compression, retrieval-augmented generation (RAG) compression, and information-preservation evaluation, this paper proposes Partitioned Information-Budgeted Prompt Compression (PIBPC). Given a textual prompt T , an effective context budget $B \leq |C|$, a uniform partition $T = (t_1, \dots, t_n)$, and a set of compatible compression algorithms parameterized by compression level, PIBPC selects one compression option for each partition so as to maximize preserved information while satisfying the budget. The model is formulated as a multiple-choice knapsack problem with optional compatibility and transition constraints. The paper also proposes an experimentally compatible set of recent text-in/text-out compressors, metrics, benchmarks, baselines, and ablations. The central claim is that prompt compression should not be treated as uniform token reduction, but as task-oriented allocation of compression pressure across prompt regions with different informational value and different risk of loss.

1 Introduction

Prompt length has become a central bottleneck for practical LLM use. Chain-of-thought prompting, in-context learning, retrieval-augmented generation, multi-turn agents, document question answering, and long-document summarization all tend to expand the

amount of text sent to the model. Survey and empirical work on prompt compression frames this expansion as both a systems problem and a modeling problem: long prompts increase memory use, latency, and API cost, but they may also contain redundant or distracting information that weakens the model’s ability to use the relevant evidence (Z. Li et al., 2025; Anonymous, 2025a; J. Xu et al., 2024).

The field has proposed several families of solutions. Hard prompt methods shorten the surface text through token pruning, lexical filtering, sentence selection, or summarization (Y. Li, 2023; Y. Li, Dong, et al., 2023; Jiang, Wu, Lin, et al., 2023; Pan et al., 2024; Jiang, Wu, Luo, et al., 2024; J. Tang et al., 2024; Y. Li, Guerin, et al., 2024). Soft prompt methods encode context into continuous representations such as gist tokens, summary vectors, memory slots, or context embeddings (Mu et al., 2023; Chevalier et al., 2023; Ge et al., 2023; Wang et al., 2024; S. Tang et al., 2024). RAG-specific methods compress retrieved documents or document embeddings before generation (F. Xu et al., 2024; Rau et al., 2024; Anonymous, 2025b). Other work emphasizes natural-language prompt encapsulation for transferability across black-box models (Chuang et al., 2024), while evaluation-focused studies show that downstream accuracy alone is insufficient because compression may lose entities, grounding, or critical details (Anonymous, 2025c; Anonymous, 2025a). Long-context management methods such as SelfExtend provide a related but distinct path by changing how models process long contexts rather than shortening the prompt (Jin et al., 2024).

Despite this progress, most compression methods are applied globally: a single method and often a single compression ratio are selected for the entire prompt. This is a mismatch with the structure of prompts. Instructions, questions, formatting constraints, retrieved evidence, examples, and background passages do not carry the same task value, nor do they tolerate the same degree of compression. A uniform compression ratio may over-compress critical information while under-compressing redundant regions.

This paper proposes Partitioned Information-Budgeted Prompt Compression (PIBPC), a generic optimization framework for selecting a compressor separately for each partition of a prompt. The framework is inspired by a grounded-theory analysis of the prompt-compression literature, whose central category was *task-oriented informational densification*: compression is not merely shortening, but the transformation of long context into a smaller representation that preserves enough usable information for the task under operational and epistemic constraints.

2 Why the Problem Is Non-Trivial

Let T be a textual prompt with length $|T|$, and let C be the context window or available context region with length $|C|$. If $|T| \leq |C|$ and the only objective is to maximize information subject to the constraint that the compressed prompt fits in C , the trivial

solution is to apply the identity compressor to all of T . This preserves all information and satisfies the constraint.

In practice, however, compression is useful even when T fits in C . Users may wish to reduce latency, lower token cost, reserve space for the model’s answer, leave room for tool outputs, add retrieved evidence, or maintain a safety margin. Therefore the relevant budget is an *effective context budget*

$$B \leq |C|, \quad (1)$$

which may be smaller than $|T|$ even if T technically fits in the model’s maximum context. The active problem is thus either $B < |T|$, or a soft-cost objective in which length is penalized.

3 Problem Setup

Let T be tokenized text. We partition it into n contiguous regions

$$T = (t_1, t_2, \dots, t_n), \quad (2)$$

where each partition has equal length:

$$|t_j| = \frac{|T|}{n}, \quad j = 1, \dots, n. \quad (3)$$

If $|T|$ is not divisible by n , the implementation uses token-balanced partitions whose sizes differ by at most one token. A sentence-aware implementation may then adjust boundaries to avoid splitting sentences, while recording the resulting size deviation.

Each partition has an estimated informational importance. In the task-aware case, given a task description, question, or instruction q , the importance of partition t_j is

$$I(t_j; q) = w_1 R(q, t_j) + w_2 H(t_j) + w_3 E(t_j) + w_4 P(t_j) + w_5 S_{\text{inst}}(t_j). \quad (4)$$

Here $R(q, t_j)$ is task relevance, $H(t_j)$ is self-information or entropy, $E(t_j)$ is entity or fact density, $P(t_j)$ is positional risk or salience, and $S_{\text{inst}}(t_j)$ identifies instructions, questions, output-format constraints, or other control text. The weights w_i may be tuned on a development set or learned from downstream performance.

4 Compression Options

Let $\mathcal{A}$ be a set of compression options. Each option is a pair

$$o = (A_k, p), \quad (5)$$

where A_k is a compression algorithm and $p \in \mathcal{P}_k$ is a compression parameter such as target retention rate, token budget, sentence budget, dynamic ratio, or number of latent slots. The identity algorithm A_0 is always included:

$$A_0(t_j) = t_j. \quad (6)$$

For each partition t_j and option o , define

$$y_{j,o} = A_o(t_j), \quad (7)$$

$$L_{j,o} = |y_{j,o}|, \quad (8)$$

$$Q_{j,o} = \text{estimated preserved informational utility of } y_{j,o}. \quad (9)$$

A generic quality model is

$$Q_{j,o} = I(t_j; q) \cdot G(t_j, y_{j,o}; q) \cdot F(t_j, y_{j,o}) - \lambda_{\text{hall}} U(t_j, y_{j,o}), \quad (10)$$

where G estimates task utility or grounding, F estimates semantic faithfulness between original and compressed text, U estimates unsupported-content or hallucination risk, and λ_{hall} controls the penalty for epistemic risk. This follows the recent shift from evaluating prompt compression only by downstream performance toward evaluating information preservation, grounding, and hallucination (Anonymous, 2025c; Anonymous, 2025a).

5 Compatibility

A compression technique is considered appropriate for PIBPC only if it satisfies two conditions. First, it must act as a function of an explicit compression parameter. Second, its output must be compatible with the outputs of the other selected techniques.

Compatibility is represented by

$$M(o, o') \in \{0, 1\}, \quad (11)$$

where $M(o, o') = 1$ means that options o and o' can coexist in the same compressed prompt. The primary experimental pool in this paper is restricted to text-in/text-out methods. This makes the outputs directly concatenable and usable with black-box LLMs

and API models, a property emphasized by natural-language and textual compression approaches (Chuang et al., 2024; F. Xu et al., 2024; Y. Li, Guerin, et al., 2024).

Soft methods such as Gist Tokens, AutoCompressors, ICAE, LLoCO, COCOM, and PISCO are highly relevant (Mu et al., 2023; Chevalier et al., 2023; Ge et al., 2023; S. Tang et al., 2024; Rau et al., 2024; Anonymous, 2025b), but they form a different compatibility class. They may depend on a specific decoder, tokenizer, adapter, embedding space, or serving pipeline. For this reason, PIBPC treats them either as a soft-only variant or as external baselines rather than mixing them with textual compressors in the same prompt string.

6 Optimization Model

Define binary decision variables

$$x_{j,o} = \begin{cases} 1, & \text{if option } o \text{ is applied to partition } t_j, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

The basic optimization problem is

$$\max_x \quad \sum_{j=1}^n \sum_{o \in \mathcal{A}} x_{j,o} Q_{j,o} \quad (13)$$

$$\text{s.t.} \quad \sum_{o \in \mathcal{A}} x_{j,o} = 1, \quad j = 1, \dots, n, \quad (14)$$

$$\sum_{j=1}^n \sum_{o \in \mathcal{A}} x_{j,o} L_{j,o} \leq B, \quad (15)$$

$$x_{j,o} + x_{j+1,o'} \leq 1, \quad \text{if } M(o, o') = 0, \quad (16)$$

$$x_{j,o} \in \{0, 1\}. \quad (17)$$

This is a multiple-choice knapsack problem with compatibility constraints. If all selected methods are textual and mutually compatible, Constraint (16) can be dropped. Without transition penalties, the problem can be solved by dynamic programming:

$$D[j, b] = \max_{o: L_{j,o} \leq b} (D[j-1, b - L_{j,o}] + Q_{j,o}). \quad (18)$$

If style or coherence discontinuity between neighboring partitions is a concern, a transition penalty can be added:

$$\max_x \sum_{j,o} x_{j,o} Q_{j,o} - \mu \sum_{j=1}^{n-1} d(o_j, o_{j+1}), \quad (19)$$

where $d(o_j, o_{j+1})$ measures a discontinuity between adjacent compression choices. This is

useful when combining token-level and sentence-level compression because token pruning may produce less fluent fragments while sentence-level methods preserve readability (Pan et al., 2024; Y. Li, Guerin, et al., 2024).

7 PIBPC Algorithm

PIBPC proceeds in five steps:

1. **Partition:** tokenize T and divide it into n balanced partitions.
2. **Score importance:** compute $I(t_j; q)$ using relevance, self-information, entity density, positional salience, and instruction priority.
3. **Generate candidates:** apply each appropriate compressor A_k with each parameter $p \in \mathcal{P}_k$ to each t_j , including A_0 .
4. **Estimate quality:** compute $L_{j,o}$ and estimate $Q_{j,o}$ using task utility, faithfulness, information preservation, grounding, and hallucination risk.
5. **Optimize:** solve Equations (13)–(17) and concatenate the selected outputs in the original order:

$$T' = y_{1,o_1} \| y_{2,o_2} \| \cdots \| y_{n,o_n}. \quad (20)$$

8 Compatible Compression Pool

Table 1 presents the proposed primary text-compatible pool. The selection favors recent, parameterized, text-in/text-out methods, so that their outputs can be concatenated into a single compressed prompt.

Table 1: Text-compatible compression pool for PIBPC.

Option	Compression parameter	pa-	Role in PIBPC	Rationale
A_0 Identity	Full retention, $p = 1$		Preserve critical partitions unchanged	Necessary to avoid damaging instructions, questions, or key evidence.
Selective Context (Y. Li, 2023; Y. Li, Dong, et al., 2023)	Retention or deletion rate based on self-information	hard	Query-agnostic compression	Simple, textual, and suitable for locally redundant regions.

Option	Compression parameter	pa-	Role in PIBPC	Rationale
LLMLingua (Jiang, Wu, Lin, et al., 2023)	Token budget, compression ratio, budget controller		Coarse-to-fine token-level compression	Strong foundational baseline for budgeted prompt compression.
LLMLingua-2 (Pan et al., 2024)	Target compression ratio or token-level keep probability		Faithful task-agnostic token classification	Recent, efficient, and designed for generalizable faithful compression.
LongLLMLingua (Jiang, Wu, Luo, et al., 2024)	Dynamic ratio, question-aware budget, document reordering		Query-aware long-context compression	Appropriate when the task question determines key evidence.
CPC (Y. Li, Guerin, et al., 2024)	Sentence budget or sentence-retention rate		Sentence-level readable compression	Preserves coherence better than aggressive token dropping.
Perception Compressor (J. Tang et al., 2024)	Dynamic ratio, open-book ratio, iterative compression level		Training-free query-guided compression	Recent method for noisy long-context scenarios.

Soft-compression methods remain important but are evaluated separately. Gist Tokens, AutoCompressors, ICAE, SoftPromptComp, LLoCO, COCOM, and PISCO motivate the broader design space of compressed representations (Mu et al., 2023; Chevalier et al., 2023; Ge et al., 2023; Wang et al., 2024; S. Tang et al., 2024; Rau et al., 2024; Anonymous, 2025b). PISCO is especially relevant as a recent RAG-oriented soft-compression baseline, but it should not be mixed with textual outputs unless the serving stack explicitly supports hybrid representations.

9 Experimental Plan

9.1 Research Questions

The experiment should answer five questions:

1. Does PIBPC outperform uniform whole-prompt compression under the same budget?
2. Does partition-level selection improve grounding and entity preservation?
3. Does the inclusion of A_0 reduce information-loss hallucination in high-importance regions?
4. When do query-aware methods outperform query-agnostic methods?
5. How much of the gain comes from the importance function, from the optimization procedure, and from algorithmic diversity?

9.2 Benchmarks

The benchmark suite should be text-only to preserve compatibility across compressors. The proposed suite is:

1. **LongBench**, for broad long-context understanding, multi-document QA, summarization, and synthetic long-context tasks, as used in recent compression evaluations (Jiang, Wu, Luo, et al., 2024; J. Tang et al., 2024; Anonymous, 2025a).
2. **ZeroSCROLLS**, for long-input generalization and task-agnostic compression evaluation, following recent LLMLingua-2 style evaluation (Pan et al., 2024).
3. **NaturalQuestions**, **TriviaQA**, and **HotpotQA**, for open-domain and multi-hop QA, enabling comparison with RAG compression and selective augmentation (F. Xu et al., 2024; Anonymous, 2025b).
4. **MuSiQue**, for compositional multi-hop reasoning in noisy long contexts, aligned with Perception Compressor’s evaluation setting (J. Tang et al., 2024).
5. **MeetingBank** or **QMSum**, for meeting summarization and question answering, where sentence-level compression can be tested against token-level alternatives.

Multimodal benchmarks should be reserved for a later phase because modality introduces additional compatibility constraints. Although empirical work has studied multimodal prompt compression effects (Anonymous, 2025a), PIBPC’s first evaluation should isolate textual compression.

9.3 Metrics

The metrics should reflect both operational and epistemic goals:

1. **Downstream performance:** Exact Match and F1 for QA, official LongBench scores, ROUGE and BERTScore for summarization.
2. **Compression:** final size $|T'|$, compression ratio, budget satisfaction, and tokens saved.
3. **Operational cost:** end-to-end latency, time to first token, estimated API cost, and memory when measurable.
4. **Information preservation:** BERTScore between original and compressed text, entity recall, evidence recall, and grounding in the source context.
5. **Hallucination:** Micro Hallucination Rate (MiHR), Macro Hallucination Rate (MaHR), Altered Semantic Hallucination, and Information Loss Hallucination, following recent empirical evaluation practice (Anonymous, 2025a; Anonymous, 2025c).
6. **Readability:** sentence-fragment rate, fluency proxy, or human readability judgments for a sample of compressed prompts.

9.4 Baselines

The evaluation should include:

1. full prompt without compression, when it fits the budget;
2. truncation from the start, middle, and end;
3. uniform Selective Context;
4. uniform LLMLingua;
5. uniform LLMLingua-2;
6. uniform LongLLMLingua;
7. uniform CPC;
8. uniform Perception Compressor;
9. PIBPC with uniform importance $I(t_j) = 1$;
10. PIBPC with one algorithm and multiple parameters;
11. full PIBPC with multiple algorithms and parameters;
12. PISCO as an external soft-compression RAG baseline when the setup allows model-specific compressed representations.

9.5 Experimental Conditions

Use relative budgets

$$B \in \{0.25|T|, 0.50|T|, 0.75|T|\}, \quad (21)$$

and, when needed, absolute budgets derived from the target LLM’s context window.

Evaluate partitions

$$n \in \{4, 8, 16\}, \quad (22)$$

and retention parameters

$$p \in \{1.0, 0.75, 0.50, 0.25, 0.125\}, \quad (23)$$

mapped to each compressor’s native control parameter.

9.6 Analysis

For each benchmark and budget, report mean performance and bootstrap confidence intervals. Use paired comparisons because each prompt instance can be evaluated under multiple compression strategies. Analyze failures qualitatively by classifying omitted evidence, semantic alteration, partition-boundary incoherence, and hallucination caused by information loss. Ablate each component of Equation (4): relevance, self-information, entity density, positional salience, and instruction priority.

10 Expected Hypotheses

The literature suggests five hypotheses:

1. PIBPC should outperform uniform compression when informational importance varies strongly across prompt regions.
2. Query-aware options should be selected more often in long-context QA, multi-hop QA, and RAG settings, where relevant evidence is sparse or position-sensitive (Jiang, Wu, Luo, et al., 2024; J. Tang et al., 2024).
3. Sentence-level methods should improve readability and reduce fragmentary prompts, whereas token-level methods should provide finer budget control (Pan et al., 2024; Y. Li, Guerin, et al., 2024).
4. The identity option A_0 should be selected for instructions, questions, and high-salience evidence, reducing hallucination due to information loss.

5. Soft RAG compression such as PISCO may be competitive in a dedicated RAG setup, but its comparison should be separated because it belongs to a different representation and compatibility class (Anonymous, 2025b).

11 Contribution

PIBPC contributes a generic decision layer above existing compressors. Rather than proposing yet another compressor, it proposes a principled way to allocate compression pressure across prompt partitions. The framework is compatible with recent findings that no single prompt-compression method dominates across all tasks, domains, ratios, and evaluation metrics (J. Xu et al., 2024; Anonymous, 2025a; Z. Li et al., 2025). It also operationalizes the idea that compression should be evaluated by preserved utility, not merely by length reduction.

12 Conclusion

Partitioned Information-Budgeted Prompt Compression formalizes prompt compression as task-oriented budget allocation. Given a prompt, a context budget, partition-level importance scores, and a compatible family of parameterized compressors, the method chooses the compression option for each partition that maximizes estimated preserved information under the budget. The framework is general enough to incorporate existing hard prompt compressors and precise enough to support a reproducible experimental plan. Its guiding principle is that the best compressed prompt is not necessarily the shortest or the most uniformly compressed one, but the one that allocates context budget where information is most valuable and loss is most dangerous.

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